

Math 520, F07
Quiz 8

Name: *Answer*
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (10 points) Convert the pair of second-order equations

$$2\frac{d^2y}{dt^2} + 2\frac{dz}{dt} + 7y = 0, \quad 5\frac{d^2z}{dt^2} + 3\frac{dy}{dt} + 3z = 0$$

into a system of 4 first-order equations for the variables

$$x_1 = y, \quad x_2 = y', \quad x_3 = z, \quad x_4 = z'.$$

And then write them in the form of $\frac{dx}{dt} = Ax$ where $x = (x_1, x_2, x_3, x_4)^T$.

Solution:

$$\begin{cases} \frac{dx_1}{dt} = x_2 \\ \frac{dx_2}{dt} = \frac{d^2y}{dt^2} = -\frac{7}{2}y - \frac{dz}{dt} = -\frac{7}{2}x_1 - x_4 \\ \frac{dx_3}{dt} = x_4 \\ \frac{dx_4}{dt} = \frac{d^2z}{dt^2} = -\frac{3}{5}\frac{dy}{dt} - \frac{3}{5}z = -\frac{3}{5}x_2 - \frac{3}{5}x_3 \end{cases}$$

Matrix Form:

$$\frac{dx}{dt} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{7}{2} & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{3}{5} & -\frac{3}{5} & 0 \end{bmatrix} x$$

Problem 2. (10 points) Show that the following set of vectors

$$\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix},$$

form a basis for $\mathbb{R}^3$.

Solution: We first need to show that $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ are linearly independent vectors.

$$c_1 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = 0$$

$$\begin{cases} -c_1 + c_2 + 2c_3 = 0 \\ c_1 - 2c_2 + c_3 = 0 \\ 2c_1 + c_2 - c_3 = 0 \end{cases} \Rightarrow \begin{cases} -c_2 + 3c_3 = 0 \\ 5c_2 - 3c_3 = 0 \end{cases} \Rightarrow \begin{cases} 4c_2 = 0 \\ c_2 = 0 \end{cases}$$

$$\Rightarrow c_3 = \frac{c_2}{3} = 0 \Rightarrow c_1 = c_2 + 2c_3 = 0$$

Thus, they are linearly independent. Again, we know $\dim(\mathbb{R}^3) = 3$. By Theorem 3 in §3.3, they form a basis for $\mathbb{R}^3$.